Waseda University
Graduate School of Information, Production and Systems

Midterm research paper

Diagnosis-Guided Augmentation and Frequency-Gated Fusion for Actor-Conditioned Human-Value Recognition in News

Peiyang Huang

Supervisor: Professor Mizuho Iwaihara

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This midterm paper develops the augmentation experiments and system analysis associated with our NTCIR-19 participant manuscript, and sets out the remaining experiments for the final thesis.

Abstract

News reports often describe actions whose implications for human values are left implicit. Recognizing these values can help analyze how individuals and institutions are portrayed through their reported conduct. Such analysis requires attention to the actor and the surrounding events, because the same report may involve people or organizations with different roles and interests. NTCIR-19 FEHU (Fact-based Event-centric Human-value Understanding) aims to evaluate systems that identify values associated with specific actors in factual news and determine whether their reported conduct aligns with or contradicts those values. This paper focuses only on Task 1, which requires value recognition at the article level for a supplied set of actors.

The study addresses limited supervision for infrequent labels, along with false positives and direction errors when expanding prediction coverage. DRA-FEHU uses error diagnosis to guide the generation of additional training examples and compares full and partial-label supervision. For inference, WAIPS-02 combines majority voting across prompts with separate direction arbitration. WAIPS-03 segments articles to form initial predictions, refines the label set using auxiliary prompt predictions and encoder scores through frequency-gated fusion, and selectively checks disagreements about value direction.

Keywords: human-value recognition; actor attribution; partial-label supervision; data augmentation; calibrated fusion; NTCIR-19 FEHU.

Contents

1 Introduction **1**
- 1.1 Background and research problem . . . 1
- 1.2 Research goal and working hypothesis . . . 1
- 1.3 Contribution and relationship to the participant paper . . . 2
- 1.4 Organization . . . 2

2 Related work **2**
- 2.1 From value recognition in arguments to actor-conditioned news analysis . . . 2
- 2.2 Synthetic supervision and diagnosis-guided augmentation . . . 3
- 2.3 Partial labels, imbalance, and the meaning of a negative example . . . 3
- 2.4 Prompt diversity, aggregation, and correlated errors . . . 3
- 2.5 Calibration and F1-based decisions . . . 4

3 Problem definition and research questions **4**
- 3.1 Input, labels, and evaluation units . . . 4
- 3.2 Learning objective and error types . . . 5
- 3.3 Research questions . . . 5

4 Diagnosis-guided augmentation and selective inference **5**
- 4.1 Overview and common data handling . . . 5

4.2 DRA-FEHU: selecting what to augment 6
4.3 Masked encoder supervision and auxiliary evidence 7
4.4 WAIPS-02: prompt majority with direction arbitration 8
4.5 WAIPS-03: granularity-controlled primary predictions 8
4.6 Support scores, frequency gating, and calibration 9
4.7 Fusion algorithm and selective direction checking 11

5 Evaluation and discussion **12**
5.1 Experimental design 12
5.2 Dataset, sample coverage, and information boundaries 12
5.3 Evaluation measures and their interpretation 13
5.4 Models, hardware, and parameter selection 14
5.5 RQ1: baseline diagnosis and the effect of representation 14
5.6 RQ2: contribution of diagnosis-guided augmentation 16
5.7 RQ3: staged fusion and the precision–recall trade-off 18
5.8 Official test results and the organizer comparison 19
5.9 Direction correction, ambiguity, and an introduced error 20
5.10 Interpretation of the results 20

6 Conclusion and work toward the final thesis **21**
6.1 Main findings and next experiments 21
6.2 Task 2 and actor-linked evidence 21

A Evidence traceability **22**

References **23**

1 Introduction

1.1 Background and research problem

News articles describe actions, intentions, and consequences involving people and institutions. A report about sanctions, for example, may discuss the security interests of one government and the harm experienced by another population. Recognizing a security-related theme is only the first part of understanding the values expressed by that report. A system must also identify the actor to whom the judgment applies and decide whether the reported conduct supports or violates the relevant value. Confusing those decisions can produce a plausible thematic label with an incorrect actor or direction.

Actor-specific value recognition supports analysis of how news portrays institutions and reports conduct affecting public welfare. The prediction concerns the action described in the article under the supplied annotation scheme; it is not a measurement of an actor's enduring moral character.

NTCIR-19 FEHU makes this problem explicit through event-centric human-value recognition [1]. Task 1 provides articles and actor identifiers. Task-1a requires coarse values for each article–actor pair, whereas Task-1b requires fine values together with alignment or contradiction. The benchmark therefore connects multilabel recognition with attribution and directional reasoning. Its hierarchy also makes the difficulty visible at two resolutions: a system can recognize the general category while confusing the fine value or its direction.

My early experiments exposed a recurring trade-off. Conservative encoder predictions covered few directed classes and missed many difficult labels. Prompt-based systems recovered more values but also generated unsupported labels or attached the wrong direction to a valid value. Increasing the candidate pool sometimes made performance worse because false positives grew faster than true positives. I therefore study how to recover missed values while distinguishing the evidence needed for inclusion from the evidence needed for direction assignment.

1.2 Research goal and working hypothesis

The goal is to improve actor-conditioned human-value recognition under a small, imbalanced training set and the official input protocol. I focus on Task 1, which requires article-level interpretation without the official subevents supplied for Task 2. My working hypothesis is that *value recovery, precision control, and direction assignment should be treated as related but separately controlled decisions*. The system should distinguish their evidence and error types, even when they share a model.

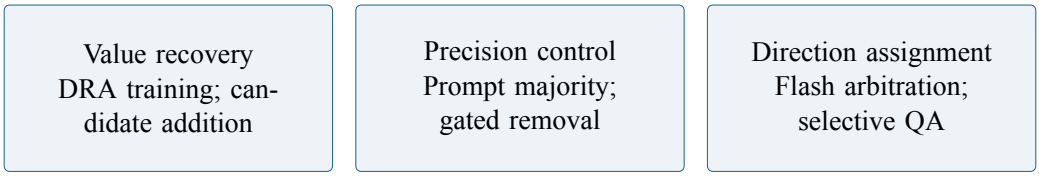


Figure 1: Three analytical roles and their associated methods. The actual fusion flow is shown in Figure 3.

1.3 Contribution and relationship to the participant paper

My work covers the design and analysis of DRA-FEHU, WAIPS-02, and the WAIPS-03/E7-G4 system. Our joint NTCIR-19 participant manuscript reports the submissions [2]. This paper develops the following methods and experiments:

1. DRA-FEHU generates examples for a frozen set of 18 diagnosed labels. Three-seed comparisons measure how augmentation and the treatment of unspecified labels affect ranking and thresholded recognition.

2. WAIPS-02 separates inclusion, decided by agreement among three prompt variants, from direction arbitration using a Flash prediction.

3. FGHF revises a segmented primary prediction through frequency-gated removal and addition, then checks direction disagreements. Saved stage outputs allow its gains and errors to be counted separately.

The augmentation experiment evaluates the encoder on its own. The fusion experiment uses a historical v2 score export whose exact correspondence to those repeated runs is incomplete; it does not isolate the effect of augmentation on the complete system.

WAIPS-01 is Liu Dongyang's label-aware ensemble and is outside my individual methodological contribution. E7 uses one historical target-label prompt from that work as an auxiliary prediction source, rather than the complete ensemble.

1.4 Organization

Section 2 reviews related work. Section 3 defines the task and research questions. Section 4 presents the methods, and Section 5 evaluates augmentation, selective fusion, and official test performance. Section 6 summarizes the completed contributions and outlines extensions for the final thesis.

2 Related work

2.1 From value recognition in arguments to actor-conditioned news analysis

Kiesel et al. [3] formulate human-value identification in arguments using 54 fine values. ValueEval evaluates 20 broader categories across multiple argument sources [4]. Their hierarchical inventories support comparison across both common concepts and less frequent distinctions. FEHU extends this setting to news containing multiple actors, where a correct prediction must identify whose action expresses a value and whether it aligns with or contradicts that value.

The broader NEVU resource organizes actor-conditioned annotations at multiple scopes, including articles and subevents [5]. FEHU uses a task-specific release with defined input and output boundaries [1]. This study uses the FEHU Task 1 split; its dataset counts should be distinguished from those of the full NEVU collection. The multi-granularity structure also motivates the Task 2 extension proposed for the final thesis.

2.2 Synthetic supervision and diagnosis-guided augmentation

Data augmentation supplies additional task supervision by constructing new examples. As reviewed by Feng et al. [6], its usefulness depends on preserving the relation between text and labels. For FEHU, this relation includes the actor and direction: preserving a news topic alone does not preserve whether an action supports or violates a value. DRA-FEHU therefore targets diagnosed directed labels and records the generation mode and supervision scope.

EAVIT trains a local value detector with explanation-based supervision and synthetic examples. The detector narrows the value definitions passed to an online model, which makes the final value judgment [7]. FGHF reverses that information flow: prompt systems provide the primary and candidate sets, while local encoder scores help revise them. DRA-FEHU additionally specifies an actor, fine value, and direction in generated news records and compares full with partial supervision. These differences define the method's scope; the experiments do not include a matched EAVIT baseline.

2.3 Partial labels, imbalance, and the meaning of a negative example

In a multilabel task, an unmentioned label is not always a verified negative. This matters when a synthetic example is generated for one target value: other values may also be expressed even if the generation request did not ask for an exhaustive annotation. Treating every omitted label as negative can introduce systematic supervision errors. Durand et al. [8] study multilabel learning with partial labels and provide a relevant precedent for distinguishing unknown entries from observed supervision. My masked objective applies that distinction to generated FEHU records.

Masking specifies which entries are supervised, whereas class weighting changes their contribution to the loss. Negative subsampling further changes the positive-to-negative balance. These mechanisms must be distinguished when interpreting the augmentation results: additional examples expand value coverage, while the supervision rule controls how their labels enter training.

The local classifier uses DeBERTaV3 [9] with low-rank adaptation, LoRA [10]. These choices support repeated local training with a compact set of trainable parameters. Evaluation includes both thresholded classification and target-label ranking, matching the encoder's roles as a recognizer and an auxiliary scorer.

2.4 Prompt diversity, aggregation, and correlated errors

Multiple model outputs provide complementary evidence. Self-consistency aggregates sampled reasoning paths [11]; WAIPS-02 instead applies three prompt formulations to the same article and requires exact directed-label agreement. WAIPS-03 combines outputs that also differ in intermediate segmentation and model configuration. These members share the article and taxonomy, so agreement is used as a support score rather than as independent observations.

Segmentation reduces competing context by replacing a full article with shorter event descriptions before prediction. Its granularity affects the evidence available to each actor: excessive fragmentation can repeat evidence, while overly broad segments can mix actions and

consequences. The self-segmentation experiments retain the article–actor output unit while varying this intermediate representation.

2.5 Calibration and F1-based decisions

Calibration relates a score to empirical prediction correctness. Guo et al. [12] distinguish classification accuracy from calibrated confidence. FGHF uses isotonic regression to map a combined support score to correctness on development candidates, fitting separate maps for removal and addition.

Lipton et al. [13] relate the F1-optimal threshold to half the optimal F1 under calibrated-probability conditions. E7 instead uses half the observed primary-system F1. This is a heuristic: the primary F1 is not the fused predictor's optimum, calibration pools different labels, and candidate decisions are constrained.

Development comparisons and official test results are reported separately, following the distinction between model selection and final assessment [14]. Calibration and evaluation use article-level grouping to keep actors from one article together. This preserves the unit of dependence when fitting folds and computing paired intervals [15].

3 Problem definition and research questions

3.1 Input, labels, and evaluation units

Let $x_i = (t_i, b_i)$ denote the title and body of article i, and let A_i be its supplied actor set. Every article has a global identifier g_i; an actor identifier $a \in A_i$ is interpreted within that article. The Task 1 input is (x_i, A_i). The output unit is $u = (g_i, a)$, not an article alone and not an actor name pooled across articles. The system must preserve the supplied identifier even when several actor names are semantically related.

Let $V = \{0, \ldots, 53\}$ be the fine, Level-1 value inventory and $C = \{0, \ldots, 19\}$ the coarse, Level-2 inventory. The official hierarchy supplies a map $h : V \to C$. The level numbers can be counterintuitive: Level 1 is the finer inventory, while Level 2 is the broader one. Let $D = \{0, 1\}$, where 0 means contradictory and 1 means aligned. Task-1b predicts a set

$$\widehat{Y}_u \subseteq V \times D. \tag{1}$$

There are 108 possible directed labels. A convenient encoder index is $k(v, d) = 2v + d$. Task-1a predictions in my systems are obtained by projection,

$$\widehat{Z}_u = \{h(v) : (v, d) \in \widehat{Y}_u\}. \tag{2}$$

Projection removes direction and merges fine values sharing a coarse category. Consequently, a direction-only correction can change Task-1b while leaving Task-1a unchanged.

The official output is a JSON list of article records, each containing its identifier and actor annotations. A fine-label annotation stores the actor identifier, Level-1 identifier, and direction. The evaluator converts these annotations to sets indexed by u, so duplicates do not increase

credit. Both directions for one fine value are permitted when the article supports them through different actions.

Task 2 uses a different unit, (g_i, s, a), where s identifies a supplied subevent. Task-2a and Task-2b use the analogous coarse and directed inventories. This paper does not claim Task 2 results. In particular, official subevents available in training and development cannot be assumed available for Task 1 test inference. A segmentation model may derive intermediate text from the test article, but it cannot read the withheld official Task 2 structure for that article.

3.2 Learning objective and error types

The learning problem is to estimate a set-valued function $f(x_i, a)$ matching the annotated values. It combines label discrimination, output cardinality, and direction assignment. The analysis distinguishes missed values, unsupported values, direction reversals, and attribution to the wrong actor. The first three are measured from prediction sets; the cases also examine how evidence relates to a named actor. Coarse F1, directed F1, and DRR capture different aspects of these errors.

3.3 Research questions

RQ1: Where does recognition fail?
How do label frequency, intermediate representation, and direction affect prompt-based and encoder-based recognition?

RQ2: What changes with augmentation?
Under a common training protocol, how does target augmentation affect label ranking and thresholded predictions? Does masking behave differently across generated-data compositions?

RQ3: What changes with selective fusion?
How do removal, addition, frequency gating, and direction checking affect the precision–recall balance of a fixed primary system?

4 Diagnosis-guided augmentation and selective inference

4.1 Overview and common data handling

The three approaches share the official article identifiers, actor identifiers, and value hierarchy. DRA-FEHU improves local-model training through targeted augmentation. WAIPS-02 aggregates prompt predictions. Frequency-gated heterogeneous fusion, FGHF, revises a primary prediction using prompt-juror support and the local encoder's continuous label scores.

All joins use identifiers rather than array position. Training articles are matched to their gold annotations by article identifier, and labels are grouped by the supplied actor identifier. The local encoder constructs one sample per labeled article–actor unit. Prompt pipelines present the actor list together with the article and ask the model to reuse those identifiers. At output time, fine-value identifiers are checked against the inventory and converted through the official

hierarchy. This common representation allows predicted sets, encoder scores, and juror votes to be aligned without comparing free-text actor names.

The prompting pipeline combines training demonstrations, the taxonomy, the article, and its actor list, then parses the model response into fine and projected coarse predictions. E1 uses two complete demonstrations and removes the early restrictions on example labels and output cardinality, allowing the model to return the values supported by the article.

4.2 DRA-FEHU: selecting what to augment

DRA-FEHU uses a frozen set T of 18 directed labels identified through baseline diagnosis. They have at most 130 official-training positives, at least three development positives, and zero recall from the official-only seed-42 baseline at threshold 0.30. Fourteen are contradictory and four aligned, spanning 12 coarse categories. The aim is to supply examples where limited training support coincides with observed recognition failure. The selected labels are listed below; the archive does not fully preserve the final selection procedure.

Target metrics apply to this development-selected subset and are reported alongside full-space metrics. In the archive's direction:value notation, T is 0:31, 0:48, 0:53, 0:16, 0:19, 0:20, 0:13, 0:37, 0:29, 0:2, 0:33, 0:32, 0:40, 0:28, 1:31, 1:10, 1:49, and 1:4.

Generation requests news-like records in which an identified actor expresses a target directed value. The first round supplies 340 examples after model-based semantic screening. The second supplies 900 records, or 50 per target, with the composition shown in Table 1. Direction contrasts distinguish support from violation; situation transformations vary the event context; hard-negative requests contrast thematic relevance with sufficient evidence.

Table 1: Second-round generation and supervision scope. The 90 hard-negative requests are a generation mode, not 90 verified negative annotations.

Generation mode	Records	Scope	Intended contrast
Targeted in-context	450	Full	Target-oriented demonstrations
Direction contrast	180	Partial	Aligned versus contradictory
Situation transformation	180	Partial	Different event context
Hard negative	90	Partial	Relevance versus sufficient evidence
Total	900	450 full; 450 partial	36 explicit known negatives

The first round received automated semantic screening; the second passed structural checks only. A six-example human pilot is too small to estimate the quality of all 1,240 records. The generation mode also differs from the observed annotation scope: only 36 second-round records carry explicit known negatives.

The combined data contains 2,468 article records and 4,575 training actor units, including 3,785 with full supervision and 790 with partial supervision. One article can contribute several actor units, so these counts describe different aspects of the training set.

4.3 Masked encoder supervision and auxiliary evidence

The local model encodes the article title and content together with an actor-conditioned text field. A classification head maps the pooled representation to 108 logits, and independent sigmoid functions produce $p_{u,\ell}$. For a training unit u, let $y_{u,\ell} \in \{0,1\}$ denote the available target and $m_{u,\ell} \in \{0,1\}$ indicate whether that target is supervised. Official records and synthetic full-multilabel records use an all-one mask. Partial records supervise their known positive and explicitly known negative entries; all other entries have zero mask.

For a minibatch B, the implemented objective is

$$\mathcal{L}(B) = \frac{\sum_{u\in B}\sum_{\ell} m_{u,\ell}\left[-w_{\ell}^{+} y_{u,\ell}\log p_{u,\ell} - (1-y_{u,\ell})\log(1-p_{u,\ell})\right]}{\sum_{u\in B}\sum_{\ell} m_{u,\ell}}. \tag{3}$$

The positive weight is computed from supervised entries,

$$w_{\ell}^{+} = \mathrm{clip}\left(\sqrt{\frac{n_{\ell}^{-}}{n_{\ell}^{+}}}, 1, 10\right), \tag{4}$$

The implementation uses $\max(n_{\ell}^{+}, 1)$ in the denominator. Positive weights affect only positive loss terms.

Masking excludes unknown synthetic labels from negative supervision. N1 and N3 further subsample supervised negatives for the selected targets, retaining approximately one or three negatives per positive before computing the loss weights. This changes the effective training balance; target precision and ranking are evaluated alongside recall.

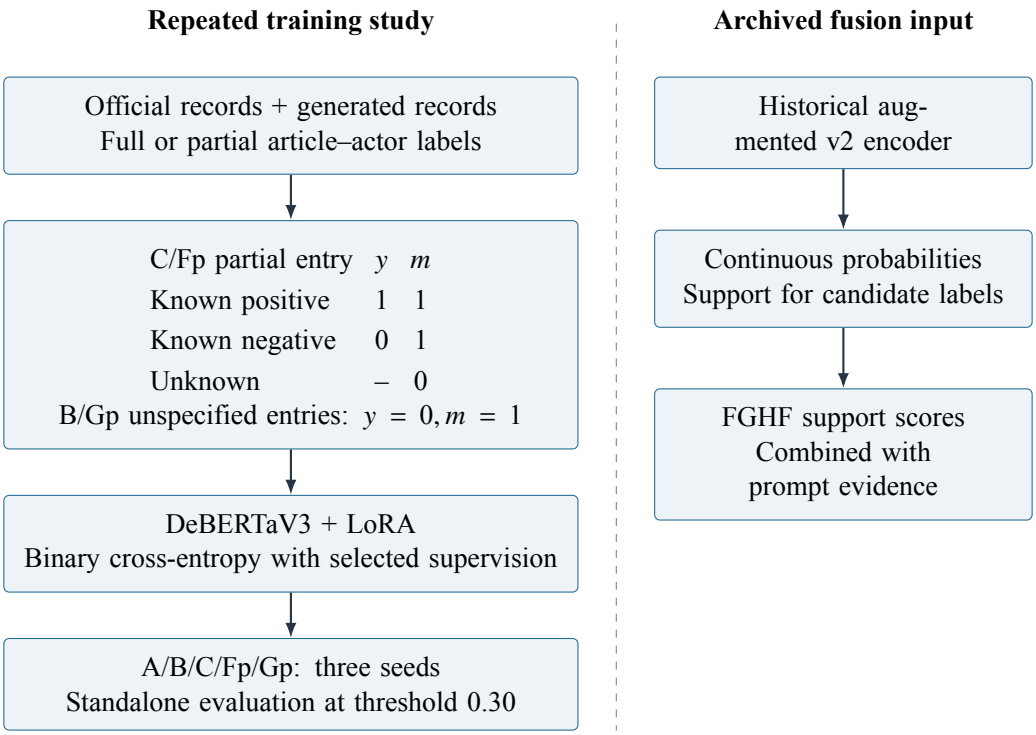


Figure 2: Encoder supervision and archived fusion scores. Masked conditions exclude unknown entries from the loss; the unmasked conditions treat unspecified labels as negative.

FGHF takes continuous encoder scores as supporting evidence for candidates supplied by the prompt systems. It uses an archived augmented v2 export, recorded separately from the repeated encoder experiments. This lets fusion use label scores without restricting its output to the encoder's fixed-threshold predictions.

4.4 WAIPS-02: prompt majority with direction arbitration

WAIPS-02 uses three Pro prompt variants: a baseline, a targeted and more balanced demonstration prompt, and a compact strict prompt. Each produces a set $S_u^{(j)} \subseteq V \times D$. The majority set is

$$M_u = \left\{ \ell : \sum_{j=1}^{3} \mathbf{1}[\ell \in S_u^{(j)}] \geq 2 \right\}. \tag{5}$$

Agreement is exact: the actor, fine value, and direction must match. A value supported in opposite directions by different members does not automatically obtain a directed majority.

A separately generated Flash prediction provides a direction lookup $q_u(v)$ for values that it recognizes. For each $(v, d) \in M_u$, WAIPS-02 retains d when the lookup is absent and otherwise uses $q_u(v)$. The result is deduplicated and projected to Task-1a. The historical implementation stores one direction per article–actor–value key; if Flash supplies both directions, the last encountered entry determines the lookup. This order-sensitive case is a limitation of the retained procedure and should be made explicit in a future revision.

Arbitration reuses the complete Flash prediction and changes direction within the majority's value coverage. On test data, 2,295 majority tuples become 2,260 final directed tuples after arbitration and set deduplication. Legal dual-direction predictions remain where the procedure retains them.

The rationale is conservative. Prompt agreement supplies the inclusion decision; a different model output supplies a directional preference. This limits the number of unsupported candidates, at the risk of losing valid labels recognized by only one prompt. Its simplicity makes WAIPS-02 an informative system-level comparison with FGHF, although their different model-call budgets mean that their scores alone do not isolate the effect of the aggregation rule.

4.5 WAIPS-03: granularity-controlled primary predictions

FGHF begins with E6b, a self-segmentation pipeline that derives shorter event descriptions from the article title and body. The segmentation prompt does not use the value taxonomy. A subsequent prediction prompt receives the derived segments and the actor inventory, then produces article-level directed values. Training demonstrations may use official training subevent descriptions, but test segmentation is generated from the Task 1 article itself.

The initial E6 configuration tends to produce finer descriptions, with a development median of 12 segments. E6b asks for approximately six to nine segments and has a median of eight. E6c reuses the E6b segmentation with a Pro non-thinking predictor. The design tests whether reducing fragmentation provides a better context for actor-conditioned judgment. The generated segments are summaries rather than guaranteed verbatim evidence spans, so their usefulness for F1 does not establish their factual completeness.

For each unit u, let P_u be the E6b primary set. Four historical jurors supply additional predictions: E1, E6, E6c, and the targeted prompt component noted in Section 1.3. The local encoder supplies $p_{u,\ell}$. These members differ in prompts, intermediate representation, or model configuration. They are heterogeneous in this practical sense, but their errors may still be strongly dependent.

4.6 Support scores, frequency gating, and calibration

Let $J_u^{(j)}$ be the set from juror j. For a candidate $\ell = (v, d)$, define

$$e_u(\ell) = \sum_j \mathbf{1}[\ell \in J_u^{(j)}], \tag{6}$$

$$r_u(v) = \sum_j \mathbf{1}[\exists d' : \ (v, d') \in J_u^{(j)}], \tag{7}$$

$$o_u(\ell) = r_u(v) - e_u(\ell). \tag{8}$$

Here o_u counts jurors supporting the value without supporting the exact directed tuple. A juror that predicts both directions contributes to e_u and r_u, not to o_u. The removal and addition scores are

$$s_{\text{rm}}(u, \ell) = e_u(\ell) + 0.25\, o_u(\ell) + \max_{d' \in D} p_{u,(v,d')}, \tag{9}$$

$$s_{\text{add}}(u, \ell) = e_u(\ell) + p_{u,\ell}. \tag{10}$$

Removal accepts cross-direction support for the same value and uses the encoder's maximum over directions. Addition uses exact directed support and the candidate's directed score.

The train-frequency gate is

$$F = \{\ell : n_\ell > 130\}. \tag{11}$$

The gate contains 38 of 108 directed labels, counted from official training data. Only these labels can be removed or added; less frequent primary tuples pass through unchanged. All 18 augmentation targets lie outside the gate, so $T \cap F = \varnothing$. Direction QA can still act on them. The cutoff was chosen during development to protect labels with little calibration support.

Removal and addition each use one isotonic regression, pooling candidates across labels. Removal candidates are primary tuples; addition candidates are juror tuples whose fine value is absent from the original primary set. The target is exact directed membership in development gold. Two folds are assigned by a deterministic hash of the article identifier, and each fold is scored using calibrators fitted on the other. Actors from the same article stay together.

The operating threshold is half the primary system's Micro-F1 on the fitting fold. Test inference uses calibrators fitted on the complete development set, with a saved threshold of approximately 0.27479, and requires no test gold.

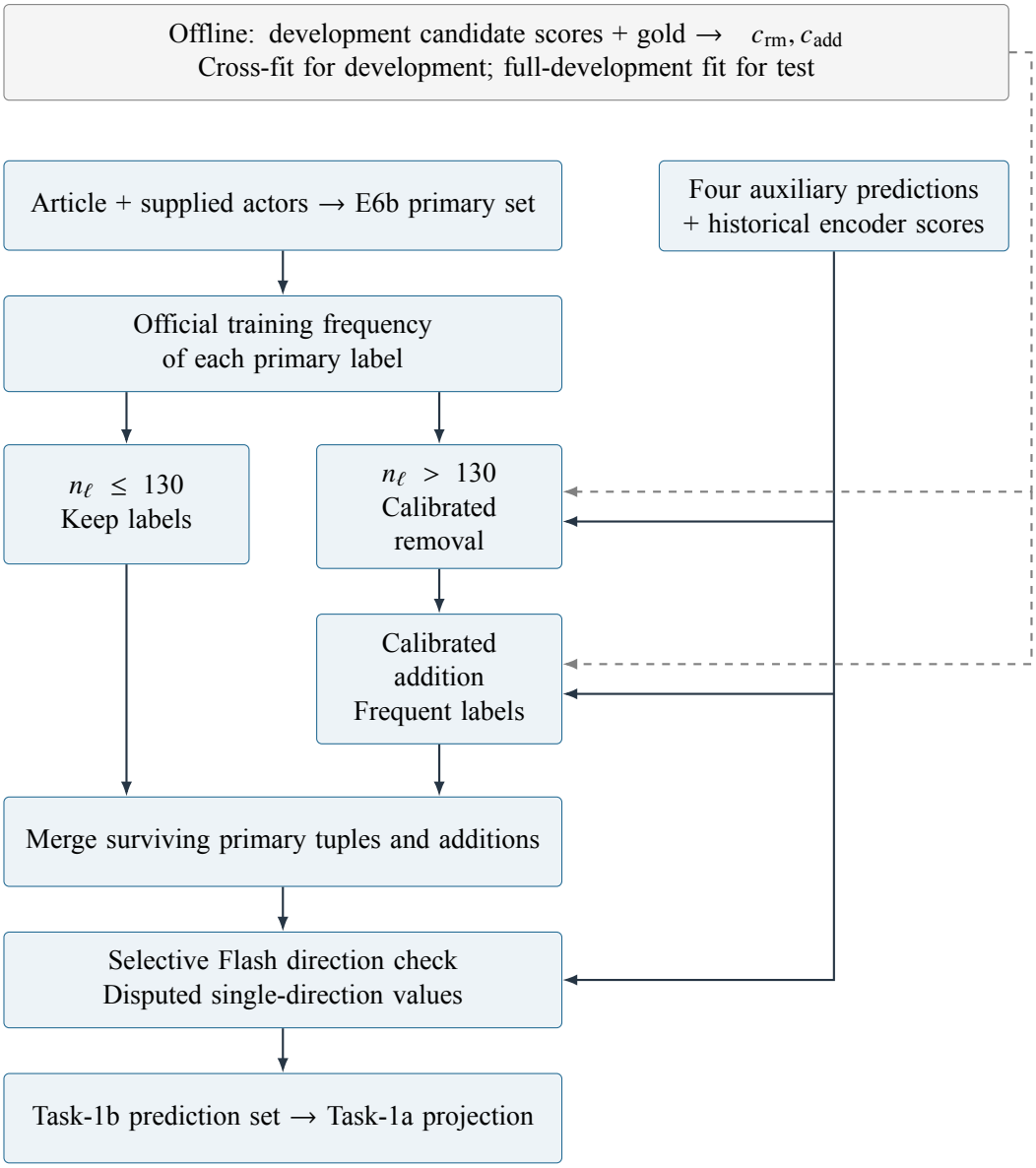


Figure 3: FGHF frequency branches and calibration. Dashed arrows supply the offline calibration maps. Removal retains at least one primary tuple; addition admits at most two directed tuples per actor.

4.7 Fusion algorithm and selective direction checking

Algorithm 1 states the retained removal and addition procedure. Low calibrated removal scores are processed first, while at least one label is retained for an actor with a nonempty primary set. Addition considers juror candidates whose value is absent after removal and admits at most two directed tuples per actor. The minimum-one constraint is a heuristic inherited from the implemented system; the task itself permits empty predictions.

Frequency-gated revision of an actor's primary set
Inputs: primary P_u; juror union C_u; frequent set F; calibrated score maps $c_{\text{rm}}, c_{\text{add}}$; threshold θ.

1. Set $R \leftarrow P_u$.
2. Rank $\ell \in R \cap F$ satisfying $c_{\text{rm}}(s_{\text{rm}}(u, \ell)) < \theta$ by ascending calibrated score.
3. For each ranked ℓ, remove it from R only if $|R| > 1$.
4. Record the surviving value set $V_R = \{v : (v, d) \in R\}$.
5. Form $B = \{(v, d) \in C_u \cap F : v \notin V_R,\ c_{\text{add}}(s_{\text{add}}(u, (v, d))) \geq \theta\}$.
6. Add the top two tuples in B by calibrated score, or all tuples if $|B| < 2$; resolve score ties deterministically as in the implementation.
7. For each single-direction value in R with an opposite-direction juror prediction, request a direction-only check. Apply a valid returned direction; retain the current direction if the recorded request fails.
8. Return the resulting set and its coarse projection from Equation (2).

Algorithm 1: The deployed decision logic. Step 6 uses the value set recorded at Step 4; it can admit both directions of a newly added value. Failure retention in Step 7 describes the historical run behavior.

Addition is applied after removal, so it can reconsider a value removed from the primary prediction. Its calibration candidates were defined using the original primary set, leaving a candidate-distribution difference to examine in later experiments. The two-tuple limit also permits both directions of a newly introduced value, consistent with the task's annotation scheme.

Direction QA is triggered when a fused single-direction value has at least one juror supporting the opposite direction. The question contains the article, actor, and value definition without revealing the proposed direction. It asks the Flash model to choose aligned or contradictory. Already dual-direction predictions are excluded from this trigger. The rationale is to spend an additional decision only where the ensemble exposes a disagreement. It does not assume that agreement proves a direction correct.

Development QA checks 176 cases: 46 directions change, 123 remain unchanged, and seven failed requests retain the current direction. Test QA checks 207 cases: 77 change, 125 remain unchanged, and five failed requests retain the current direction. QA preserves value identity and the Task-1a projection. Section 5 examines direction-check outcomes.

5 Evaluation and discussion

5.1 Experimental design

The evaluation compares baseline representations, repeated encoder training, and successive fusion stages. Encoder experiments use seeds 42, 43, and 44. Prompt and fusion results use saved predictions; official test scores come from the task overview and participant manuscript. The appendix maps these results to their artifacts.

5.2 Dataset, sample coverage, and information boundaries

The study uses the official FEHU release retained in the project. Table 2 distinguishes articles, supplied actors, and actors with positive gold annotations. The local encoder loader historically constructs 3,335 training and 456 development units from labeled actors. The full development event base supplies 459 actors. These are different coverage conventions, even though the three additional development actors have no positive article-level gold labels. A predictor can still incur false positives on such actors. Encoder and prompt comparisons must retain that distinction.

Table 2: Counts from the retained event-base and gold files. Gold actors count article–actor units with positive article-level annotations. The two test inputs contain different articles; Task 2 test subevents are not a hidden resource for Task 1.

Split/input	Articles	Supplied actors	Gold actors	Subevents
Training	1,228	3,381	3,335	10,822
Development	177	459	456	1,526
Task 1 test	175	479	Withheld	Not supplied
Task 2 test	175	444	Withheld	1,616

The training set has 16,905 unique directed article–actor labels and 13,371 coarse labels after projection. Development has 2,390 directed labels and 1,861 coarse labels. Ninety-three directed classes have positive development support. Article identifiers are disjoint across training, development, Task 1 test, and Task 2 test. This identifier check establishes that the split files do not reuse the same article records. It does not establish semantic independence between news stories or rule out related events across splits.

Figure 4 relates the long-tailed distribution to augmentation and fusion. Training support ranges from zero to 1,735 positive article–actor units. Three directed labels have no positives. The 18 augmentation targets fall in the tail, beyond the 38 labels eligible for removal and addition. The 16,905 tuples share 1,228 articles and should not be treated as independent documents.

The experiments use the corrected actor-aligned training and development release. Reported system comparisons are tied to the saved predictions, metrics, and configurations listed in the appendix.

Few-shot examples are drawn from training. Target selection, early stopping, prompt choices, calibration, and fusion selection use development information. Official Task 1 test labels are not available locally. Experiments that use official development subevents, such as

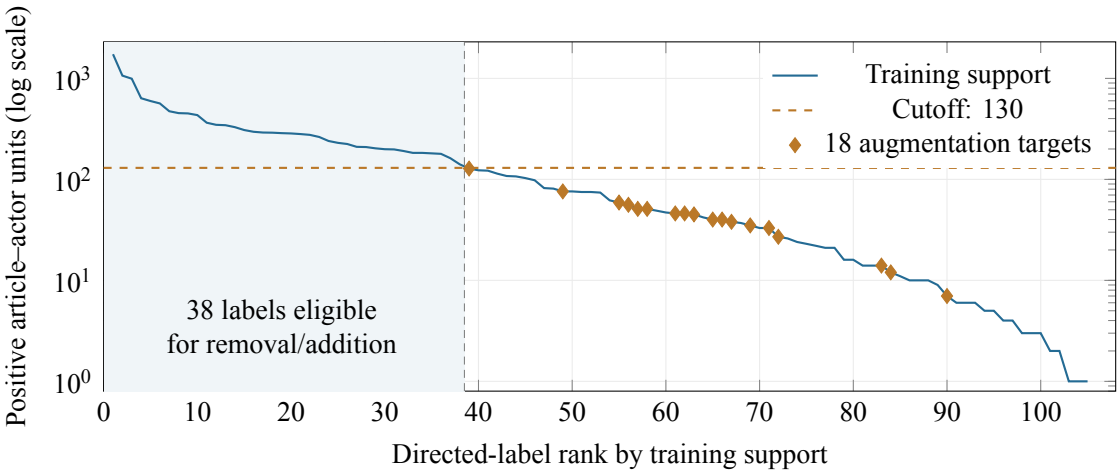


Figure 4: Training frequencies, with the 38-label gate shaded and augmentation targets marked. Three zero-frequency labels are omitted from the logarithmic curve. All targets lie outside the gate.

the earlier E3/E3a diagnostics, have extra intermediate information compared with Task 1 test inference and are excluded from the main directly applicable system table. Generated self-segmentation is allowed because it uses only the supplied article at inference time.

5.3 Evaluation measures and their interpretation

For the union U of gold and predicted instance keys, define $\mathrm{TP} = \sum_u |Y_u \cap \widehat{Y}_u|$, $\mathrm{FP} = \sum_u |\widehat{Y}_u \setminus Y_u|$, and $\mathrm{FN} = \sum_u |Y_u \setminus \widehat{Y}_u|$. Micro-F1 is

$$F_{1,\mathrm{micro}} = \frac{2\mathrm{TP}}{2\mathrm{TP} + \mathrm{FP} + \mathrm{FN}}. \tag{12}$$

The official macro score averages per-label F1 over the labels with positive gold support in the specified label universe:

$$F^{G}_{1,\mathrm{macro}} = \frac{1}{|L_G|} \sum_{\ell \in L_G} \frac{2\mathrm{TP}_\ell}{2\mathrm{TP}_\ell + \mathrm{FP}_\ell + \mathrm{FN}_\ell}, \qquad L_G = \{\ell : \mathrm{TP}_\ell + \mathrm{FN}_\ell > 0\}. \tag{13}$$

It is not the harmonic mean of macro-averaged precision and recall. Predictions for a label absent from gold can affect micro precision while receiving no separate term in gold-supported Macro-F1. Both metrics are needed to characterize a system's behavior.

For direction, let $E = \{(u, v) : |\{d : (v, d) \in Y_u\}| = 1\}$ and let $d^*_{u,v}$ be the unique gold direction. The evaluator computes

$$\mathrm{DRR} = \frac{\sum_{(u,v)\in E} \mathbf{1}[(v, 1 - d^*_{u,v}) \in \widehat{Y}_u]}{|E|}. \tag{14}$$

Development contains 2,222 such gold-exclusive pairs. Another 84 actor–value pairs have both gold directions and are excluded from DRR. If a prediction contains both directions for an exclusive gold value, it still counts as a reversal. A missing prediction does not count as a

reversal. Thus an empty system has zero DRR but no useful recall; DRR must always be read alongside F1 or coverage.

The scorer deduplicates labels, treats missing predictions as empty sets, and counts additional predicted instance keys as false positives. Identifier and schema checks are performed separately during export.

For the 18 selected targets, I additionally report the mean per-label ROC-AUC from retained probability analyses. This measures positive–negative ranking over the evaluated actor units; it is not final multilabel F1 and does not determine a deployable threshold. Thresholds chosen separately using each target's development gold would provide an oracle diagnostic rather than an unbiased test result. The repeated-run plots therefore show ranking and thresholded output as different views of the same model.

5.4 Models, hardware, and parameter selection

The local encoder is `microsoft/deberta-v3-base`, with 12 layers and hidden width 768. Its official model card reports 86 million backbone parameters and 98 million embedding parameters [16]. The retained implementation adds a 108-output linear head and applies LoRA to query and value projections. A preserved smoke-run configuration reports 589,824 trainable adapter parameters; this count does not include every newly added head parameter.

The external models are recorded as `deepseek-v4-pro` and `deepseek-v4-flash`; augmentation uses the Pro preview described in the experiment records. The official organizer comparison is a separately adapted Llama-3.1-8B-Instruct system [1].

The archived smoke run used an NVIDIA GeForce RTX 4090. Its software environment comprised Python 3.11.15, PyTorch 2.6.0 with CUDA 12.4, Transformers 5.8.1, and PEFT 0.19.1. These settings document that smoke run; complete environment records were not retained for every formal experiment.

WAIPS-02 uses four article passes at test time. E7 uses two segmentation passes, five prediction passes, and additional checks for direction disagreements. It therefore requires more requests, but their different lengths prevent a fair cost comparison without token, retry, and latency measurements.

5.5 RQ1: baseline diagnosis and the effect of representation

Table 4 groups the saved development outputs by system type. The official-only seed-42 encoder has directed Macro-F1 0.0305; the historical augmented encoder reaches 0.1303 but has higher DRR. RQ2 examines augmentation and its supervision settings through repeated training.

Self-segmentation affects different metrics differently. E6 improves directed Macro-F1 over E1 but lowers Micro-F1 and raises DRR. E6b's granularity control improves both directed F1 measures over E6 and reduces DRR, although DRR remains above E1. E6c obtains stronger coarse scores but lower directed scores than E6b. This pattern suggests that identifying relevant value categories and choosing their fine labels and directions place different demands on the predictor.

Table 3: Principal implementation settings for the reported experiments.

Component	Recorded setting	How it is used or selected
Encoder input	Maximum 512 tokens; article and actor pair	Truncation limits accessible article evidence
LoRA	Rank 16; scaling 32; dropout 0.1	Query/value projection adaptation
Optimization	AdamW; learning rate 2×10^{-4}; weight decay 0.01	Cosine schedule; warmup fraction 0.1
Training schedule	Batch 8; accumulation 4; at most 15 epochs	Patience 4; best development Micro-F1
Prediction threshold	Global 0.30	Common standalone encoder operating point
Repeated runs	Seeds 42, 43, 44	Same three seeds across conditions
E1 prompting	Two complete training examples; temperature 0.1	Prompt repair removes example/output caps
WAIPS-02	Three Pro prompts plus one Flash prediction; temperature 0	Exact majority and cached direction arbitration
FGHF gate and scores	Count > 130; value-only weight 0.25; at most 2 additions	Train-frequency gate; development-designed rules
FGHF calibration	Two article folds; isotonic maps; $F_1/2$ threshold	Cross-fit on development; full-development fit for test
Direction QA	Flash; temperature 0.1; 300 output tokens; up to 3 attempts	Only explicit single-direction disagreements
E6/E6b; E6c	Flash; Pro, respectively; temperature 0.1	E6c reuses E6b segments and explicitly disables thinking

Table 4: Development results on 177 articles. Encoder rows follow the historical 456 labeled-actor convention; prompt and fusion files are checked against the 459 supplied actors. Scores are single retained outputs, not repeated external-model means. DRR is lower-is-better; F1 is higher-is-better.

System	1a Micro	1a Macro	1b Micro	1b Macro	DRR ↓
Standalone encoders (456 labeled actors)					
DeBERTa official-only, seed 42	0.5090	0.1351	0.3023	0.0305	0.2165
DeBERTa augmented v2, historical	0.5405	0.3420	0.3306	0.1303	0.2880
Prompt systems (459 supplied actors)					
E1 repaired few-shot prompt	0.6574	0.5018	0.5480	0.3719	0.0347
E6 self-segmentation	0.6535	0.5260	0.5412	0.4079	0.0504
E6b controlled segmentation	0.6621	0.5286	0.5496	0.4124	0.0459
E6c Pro with E6b segments	0.6774	0.5600	0.5343	0.3725	0.0657
WAIPS-02 selected dev policy	0.6840	0.5573	0.5560	0.3816	0.0338
Fusion system					
E7-G4 complete fusion	0.6874	0.5471	0.5817	0.4308	0.0419

For RQ1, encoder recognition is limited by narrow directed-label coverage, while prompt systems trade higher coverage against unsupported labels and reversals. E6b's coarser segmentation improves its directed scores over E6, but the saved generations do not establish how often segmentation repairs actor attribution. That question requires evidence annotations.

5.6 RQ2: contribution of diagnosis-guided augmentation

The encoder comparison isolates the change from official-only to augmented training and examines the supervision rules. A uses official data only. B adds 900 second-round records with unspecified labels treated as negative; C uses the same records with masks. Fp combines the 340 first-round and 900 second-round records with masks, while Gp treats unspecified entries in that combined set as negative. Table 5 reports means and sample standard deviations from seeds 42, 43, and 44, computed from the saved per-run metrics.

Table 5: Encoder development results, seeds 42/43/44. SD is the sample standard deviation of saved four-decimal run metrics, not a confidence interval. DRR and target Micro-F1 show seed means; AUC averages the 18 target labels within each run.

Cell	1b Micro ± SD	1b Macro ± SD	Target AUC ± SD	DRR	Target F1
A	0.3008 ± 0.0037	0.0284 ± 0.0018	0.5133 ± 0.0304	0.1580	0.0000
B	0.3109 ± 0.0043	0.0890 ± 0.0370	0.6242 ± 0.0108	0.2823	0.0319
C	0.3099 ± 0.0168	0.0470 ± 0.0180	0.5770 ± 0.0282	0.2196	0.0081
Fp	0.3211 ± 0.0346	0.0878 ± 0.0838	0.6318 ± 0.1065	0.1772	0.0447
Gp	0.3030 ± 0.0026	0.0779 ± 0.0126	0.5981 ± 0.0210	0.2850	0.0260
N3	0.2347 ± 0.0675	0.1074 ± 0.0663	0.6079 ± 0.1205	0.3188	0.0537
N1	0.1434 ± 0.0191	0.0424 ± 0.0040	0.5066 ± 0.0494	0.2975	0.0351

B improves full-space Macro-F1 and target AUC over A in all three seeds. Its mean target Micro-F1 is nevertheless only 0.0319 at threshold 0.30. The augmentation improves label ranking, but converting those scores into reliable target predictions remains difficult. Macro-F1 and target F1 therefore provide different evidence about the benefit.

On the same 900 records, masking lowers target AUC in all three seeds: C has mean Macro-F1 0.0470 versus B's 0.0890, although its DRR is lower. With both rounds, Fp has higher mean Macro-F1 than Gp, but a standard deviation of 0.0838, and two seeds do not correctly predict any target positives. Figure 5 shows that the higher mean is driven by one run. Masking's value therefore depends on the data composition, and stability remains a central issue.

Negative subsampling substantially increases target recall but leaves precision very low (Table 6). Many additional predictions are false positives, so the higher recall does not reliably improve target F1. Reducing negative supervision alone does not resolve target recognition at a fixed threshold.

The RQ2 results support adding these generated records under the tested protocol, particularly condition B. They do not separate the effect of diagnosis-guided selection from equal-sized generic generation or oversampling. Checkpoints were selected by development Micro-F1 to

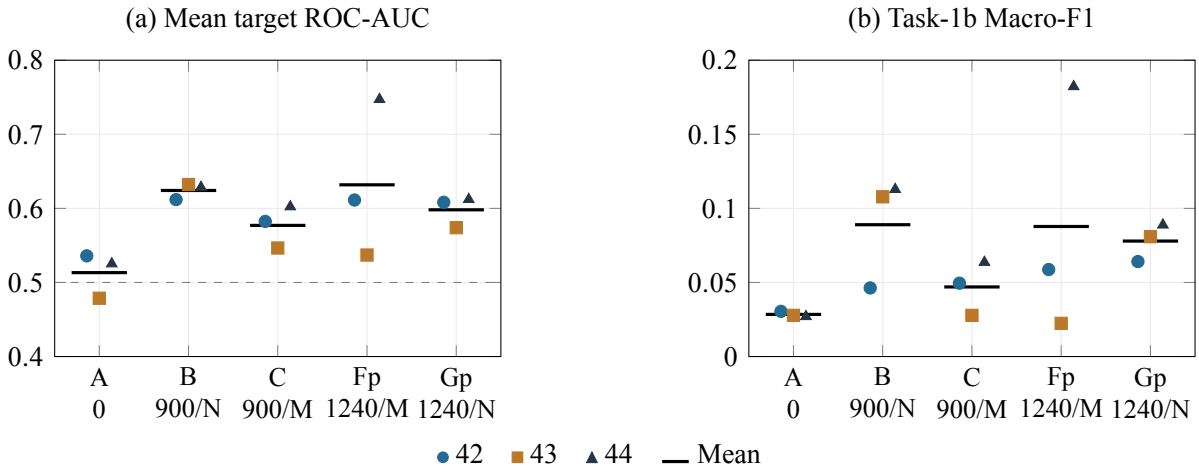


Figure 5: Three seeds and their means. Tick labels show added records: N treats unspecified labels as negative; M masks them. The AUC reference is 0.5. Fp variability is visible in both panels; panel scales differ.

Table 6: Target results with negative subsampling. All entries are three-seed means; precision, recall, and F1 are calculated per run before averaging, with zero precision for runs without target predictions. Each run has 140 target gold positives.

Cell	Target TP	Target FP	Precision	Recall	Target F1	AUC
Fp	12.3	125.0	0.0299	0.0881	0.0447	0.6318
N3	83.0	3,319.0	0.0284	0.5929	0.0537	0.6079
N1	131.3	7,255.3	0.0179	0.9381	0.0351	0.5066

keep a common training protocol, although Macro-F1 and target ranking are the main diagnostic measures. A separate selection split is needed to examine that choice and threshold sensitivity.

5.7 RQ3: staged fusion and the precision–recall trade-off

The E7 sequence starts from the same E6b primary predictions and applies successive interventions. Table 7 is the central component comparison. G1 removes weak frequent-label predictions. G3 adds supported candidates after removal. G4 applies selective direction QA to G3. The two diagnostic branches, G2 and G5, change a juror or the frequency gate within the removal-only configuration. They should be compared with G1, not treated as complete G4 ablations.

Table 7: Staged comparison and removal-branch controls on 177 development articles. G0: E6b; G1: removal; G3: removal and addition; G4: direction QA; G2: no target-prompt juror; G5: no gate. P/R are micro precision/recall.

Cell	1b P	1b R	1b Micro	1b Macro	DRR	1a Micro	Tuples
G0	0.5017	0.6075	0.5496	0.4124	0.0459	0.6621	2,894
G1	0.5585	0.5632	0.5608	0.4146	0.0374	0.6587	2,410
G3	0.5389	0.6255	0.5790	0.4266	0.0450	0.6874	2,774
G4	0.5415	0.6285	0.5817	0.4308	0.0419	0.6874	2,774
Removal-only branches, compared with G1							
G2	0.5614	0.5623	0.5619	0.4135	0.0369	0.6582	2,394
G5	0.5762	0.5644	0.5703	0.3900	0.0410	0.6707	2,341

Removal aims to filter frequent-label predictions with weak combined support. It raises precision at the cost of some recall, improving Micro-F1 from 0.5496 to 0.5608. Addition then recovers labels supported by the auxiliary predictions and encoder, raising Micro-F1 to 0.5790 while increasing recall. The two stages address different errors: removal controls unsupported predictions, while addition addresses omissions.

Direction QA addresses disagreements that remain after label selection. It raises Micro-F1 to 0.5817 and reduces DRR from 0.0450 to 0.0419, without changing the coarse predictions. The saved decisions contain both corrections and introduced errors, so its benefit is a small net improvement rather than uniformly better direction judgments. The cases below illustrate both outcomes.

Frequency gating supports macro coverage. Removal-only G1 has Macro-F1 0.4146 versus 0.3900 for ungated G5, while G5 has higher Micro-F1 (0.5703 versus 0.5608). The gate protects less-supported classes at this operating point. G2, which removes the targeted-prompt juror from the removal branch, gives a similar micro–macro trade-off.

The selected targets lie outside the frequency gate, so removal and addition preserve their primary predictions. Their Micro-F1 changes only at the QA stage, from 0.3145 to 0.3264. This is the intended role of the gate: low-frequency labels remain available even when calibration support is limited.

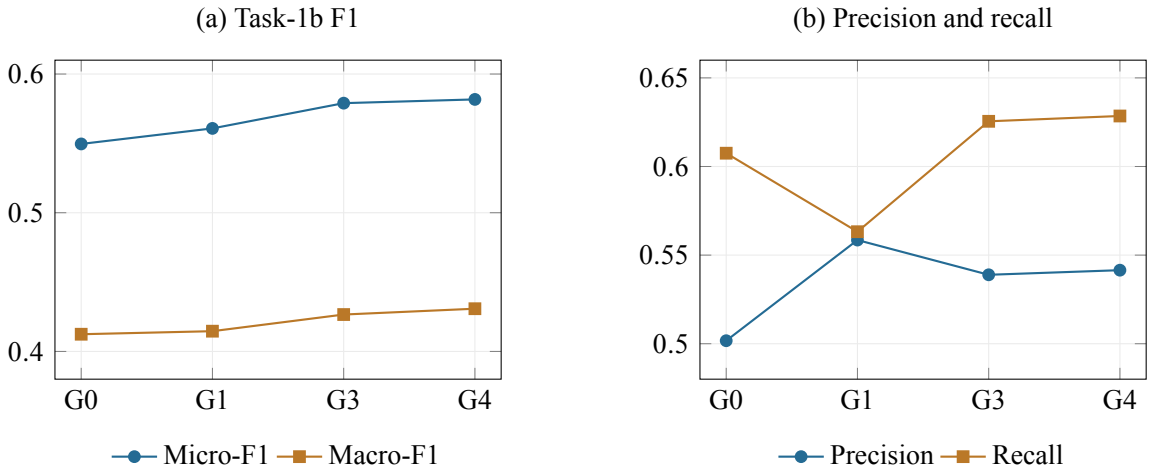


Figure 6: Gains and trade-offs across saved stages. The 95% interval [0.0209, 0.0433] applies only to the overall G4-minus-G0 Micro-F1 comparison.

An article-level paired bootstrap with 10,000 resamples gives an overall G4-minus-E6b Micro-F1 difference of 0.0321 and a 95% interval of [0.0209, 0.0433]. It keeps each article's actors together and compares the saved outputs; it does not measure variation from retraining or selecting another system.

5.8 Official test results and the organizer comparison

Table 8 reproduces the official Task 1 results reported in the retained NTCIR overview and participant paper [1, 2]. These are official scores for the submitted predictions, not evaluations newly run against local test labels. WAIPS-02 corresponds to the DS_final majority-and-arbitration submission. WAIPS-03 corresponds to the E7-G4 fusion submission.

Table 8: Official Task 1 test results: 175 articles, 479 supplied actors. All four participant submissions are shown; the organizer is listed separately. Primary ranking uses Micro-F1.

	Task 1a		Task 1b		
System	Micro-F1	Macro-F1	Micro-F1	Macro-F1	DRR ↓
WAIPS-01	0.6589	0.5366	0.5543	0.3277	0.0345
WAIPS-02	0.6930	0.5646	0.5496	0.3580	0.0395
WAIPS-03	0.6862	0.5705	0.5808	0.3952	0.0477
IIUoT-01	0.7042	0.5808	0.5979	0.3930	0.0613
Organizer (unranked)	0.6782	0.5405	0.6284	0.4402	0.0386

WAIPS-03 has higher directed Micro-F1 and Macro-F1 than WAIPS-02, alongside higher DRR and slightly lower coarse Micro-F1. Its Task-1b Macro-F1 is numerically highest among the four participant submissions. IIUoT-01 leads on Micro-F1, the official primary ranking metric; the organizer baseline is higher on both directed F1 measures.

The organizer comparison suggests a useful next baseline. Its reported Llama-3.1-8B-Instruct system uses LoRA, a 4,096-token context, and article plus subevent training supervi-

sion. Its configuration differs from my 512-token DeBERTa classifier in backbone, context length, objective, and training examples. The stronger score is evidence that a locally adapted model can be competitive in this task. It does not identify which of those differences caused the advantage. A matched article-only versus article-plus-subevent comparison is needed before claiming that subevent supervision explains the gap.

WAIPS-03's development and test Micro-F1 are close, at 0.5817 and 0.5808, while Macro-F1 falls from 0.4308 to 0.3952. Test labels are unavailable locally, so neither a test interval nor a label-level account of that macro decrease can be computed here.

5.9 Direction correction, ambiguity, and an introduced error

Table 9 gives three development cases with source excerpts. The excerpts are statements reported in the news; the gold interpretation is a separate annotation. They show how the same direction-checking procedure can correct an error, switch between two supported directions, or introduce an error.

Table 9: Selected direction checks. Value 27 is stable society; value 26 is have a safe country. Actor suffixes complete the article identifier. Quotes are from the news text, not gold explanations.

Case and decision	Reported evidence and annotation comparison
Venezuela sanctions `0-4491-3-3`, actor -0 Value 27: 1 → 0	For *US Government and Allied Nations*, the article quotes Uzcátegui: "The situation will deteriorate with these last sanctions especially." This is the interviewee's forecast. Gold links sanctions to instability and assigns only contradiction; the flip corrects the directed tuple.
Paris police `0-4403-2-2`, actor -1 Value 27: 0 → 1	For *French Authorities and Police*, the report includes "There was a failing" and a proposal that concerns "ought to trigger an 'automatic alert'." Gold annotates both security failure and efforts to restore order. Both predicted directions match gold; the flip adds no TP and the pair is excluded from DRR.
Chile protests `0-4716-3-1`, actor -1 Value 26: 1 → 0	For *Government and Security Forces*: "He justified the decision, saying he wanted to maintain public order against vandals and crime." This reports the government's stated purpose. Gold assigns alignment; QA changes a correct direction to a wrong one.

The Chile report also discusses force and allegations of abuse, but the saved QA decision contains no rationale establishing why the model flipped. The cases show why future evidence evaluation must distinguish a reported intention, an action, and its consequences for a specified actor. A label alone cannot show which passage determined the direction.

5.10 Interpretation of the results

The experiments support two findings: augmentation improves the local encoder under specific supervision settings, and selective fusion improves the submitted system. Their causal connection has not yet been isolated. Development data also guided target selection and system choices, so calibration cross-fitting does not remove all selection bias [14]. The official test assesses the submitted systems separately. For the final thesis, the main gaps are a direct

augmentation-to-fusion comparison, more stable masked training, and fuller preservation of the target-selection and historical v2 records.

6 Conclusion and work toward the final thesis

6.1 Main findings and next experiments

Generated records improve encoder Macro-F1 and target ranking, with the most consistent result in the 900-record condition that treats unspecified labels as negative. Masking has different effects across data compositions and remains unstable with the combined data. At system level, removal and addition improve the balance between precision and recall, while direction checks provide a smaller final gain.

The next experiment will fix the primary and auxiliary predictions and compare fusion with no encoder scores, official-only scores, and augmented scores under the same calibration protocol. Equal-sized generic generation and official-example oversampling will test whether diagnosis-guided selection adds value beyond extra supervision. A separate selection split and a sample audit of generated labels will support the study of masking, thresholds, and training stability. The full-system comparison will also include calibration quality and inference cost.

6.2 Task 2 and actor-linked evidence

Task 2 predicts values for article–subevent–actor tuples. Its supplied subevents make it possible to test whether more local supervision helps distinguish actors and opposing directions within an article.

I will compare article-only with article-plus-subevent supervision using the same model, context length, and training budget, starting with an 8B instruction-model baseline. All instances from one article will share a fold. Task 1 inference will retain its article-only input protocol, while Task 2 uses supplied subevents. Aggregation will retain the full article because its labels need not equal the union of local predictions.

A complementary study will link each predicted value and direction to the actor, reported action, and supporting passage. Evidence annotations, including cases with opposing actions, will let us assess whether the cited text supports that relation. Evaluation will report evidence correctness, coarse and directed F1, DRR, actor coverage, and inference cost.

A Evidence traceability

Table 10 maps results to archived artifacts. Saved outputs and configurations identify historical runs, since some source files have subsequently changed.

Table 10: Run identity and evidence status. Historical V2 is not an identified R2/R3 checkpoint.

ID	Result or role	Available evidence and limits
R2 / R3	A/B/C; Fp/Gp/N1/N3	Three-seed metrics and launch settings; not every checkpoint retained
T18	Target selection	Explicit target list and baseline probabilities; 39-to-18 rule missing
V2	Historical encoder input to E7	Frozen predictions and probability hash; original probability file and checkpoint lineage incomplete
S6	E6/E6b/E6c predictions	Saved configurations, segments, and outputs
S7	G0/G1/G3/G4; G2/G5 controls	Predictions, metrics, actions, QA decisions, bootstrap
M2 / M3	WAIPS-02 / WAIPS-03 test	Submission manifests and prediction sets
OFF	Official scores	Task overview and participant manuscript; no local test gold

WAIPS-02 predicts for all 175 Task 1 test articles and 479 actors. WAIPS-03 covers the same articles with nonempty predictions for 470 actors, producing 2,764 directed and 2,136 coarse tuples. Both submissions pass identifier, duplicate, and projection checks. Missing actor predictions are evaluated as empty sets; these format checks do not reproduce model training or validate semantic correctness.

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